Visual Composition

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Visual Composition before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Image Generation Models

Without opening the old lesson, explain one key idea from Image Generation Models. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Structured Output and Schemas

Without opening the old lesson, explain one key idea from Structured **Output** and Schemas. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Prompt as a Contract

Without opening the old lesson, explain one key idea from Prompt as a Contract. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Visual Composition trace and record the first step you can explain without help.

Use W2: Trace the worked case for Image generation and **provenance**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Visual Composition explanation against the supplied facts before discussing it.

Use the worked case for Image generation and **provenance**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Changed independent case

A supplied drawing description says three stars are above a triangle, while a circle crosses the edge of a rectangle. The brief requires three stars above the triangle and the circle fully inside the rectangle.

Evaluate this description.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Visual Composition target to a trace and a justified conclusion.

An image brief requires two trees, a bench between them and the sun upper-right. A result has all objects but the bench is left of both trees. Write a spatial test that detects this.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Visual Composition case and show the deciding evidence.

Brief: large tree left, small bicycle right, open sky above. **Output** puts bicycle left.

Create a visual checklist.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

WORKBOOK · 7 OF 21

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

A poster puts pale yellow labels on white and hides the main object behind decorations. Suggest two revisions and how to check them.

My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 11 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Visual Composition adds to Verified Study Buddy and keep the evidence.

Create age-fit explanatory visuals with labeled purpose and no invented documentary **evidence**.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

An image brief requires two trees, a bench between them and the sun upper-right. A result has all objects but the bench is left of both trees. Write a spatial test that detects this.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Visual Composition facts and keep your reasoning.

Where to find the supplied example

Reference for this exercise: open this week's student lesson, "Visual Composition", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Trace the worked case

> Trace the worked case. Apply the supplied Visual Composition facts and keep your reasoning.

A fictional poster requires exactly three labelled circles. Generated image shows four circles and one unreadable label. A caption calls it a photograph of a real lesson.

Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Audit the limit

> Audit the limit. Apply the supplied Visual Composition facts and keep your reasoning.

A report omits this qualification: **Provenance** tracks production history; it does not prove the content is accurate or establish every legal permission.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Visual Composition facts and keep your reasoning.

A generated diagram requires two arrows and labels **input**/**output**. It has two arrows but swapped labels. Metadata records generation, yet the caption says experimental **observation**. Audit.

> **My reasoning and evidence**

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Visual Composition facts and keep your reasoning.

> **Where to find the supplied example**
>
> Reference for this exercise: open this week's student lesson, "Visual Composition", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Visual Composition

> A fictional Study Buddy requests a science-poster layout, then audits a supplied drawing description. Explain the labelled step before checking the changed condition.

A fictional Study Buddy requests a science-poster layout, then audits a supplied drawing description.

Brief: exactly two labelled bottles left of one beaker; one arrow from bottles toward beaker.

Supplied drawing description: two labelled bottles are right of a beaker; an arrow points from beaker toward bottles.

AI ACADEMY / AI-3/11

Visual Composition

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Count objects

Two bottles and one beaker satisfy the count requirement.

2. Check position

Bottles on the right fail the required left-of relation.

3. Check direction

Arrow from beaker to bottles reverses the required flow.

4. Repair brief

Specify bottles-left, beaker-right, arrow left-to-right; verify the revised output.

CHANGE ONE THING

Move bottles left of the beaker but keep the arrow pointing from beaker toward bottles.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Which checks pass after that single change, and what still needs repair?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Create Study Buddy diagram rubrics covering counts, readable labels, containment, order and meaningful arrow direction.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Verified Study Buddy record.

Project: Verified Study Buddy

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Create Study Buddy diagram rubrics covering counts, readable labels, containment, order and meaningful arrow direction.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional Study Buddy requests a science-poster layout, then audits a supplied drawing description.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Specify bottles-left, beaker-right, arrow left-to-right; verify the revised **output**.

> 1. Check position

2. Repair brief

3. Count objects

4. Check direction

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Repair brief Specify bottles-left, beaker-right, arrow left-to-right; verify the revised **output**.

Having every requested object is not the same as conveying the requested **relationship**. Position and arrow direction can change a scientific explanation while object counts remain correct.

Audit the provided description honestly: no actual image was supplied, so this evaluation concerns its stated layout rather than a claimed visual inspection.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Move bottles left of the beaker but keep the arrow pointing from beaker toward bottles. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. Position now passes; arrow direction still fails.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Position now passes; arrow direction still fails. Changing object placement does not automatically reverse an arrow's semantic source and destination.

Repair a missing building block

Review Image Generation Models (ai-3/10). Goal: Explain the **mechanism** and terminology of image generation and **provenance**.

Short practice: What may these two grids explain, and what can they not establish about a real generator?. Return to this week and explain what you changed.

Review Structured **Output** and Schemas (ai-3/8). Goal: Separate JSON parsing, schema validation and source support checks. Short practice: Which object demonstrates that valid JSON and valid types are insufficient for truth?.

Return to this week and explain what you changed.

Review Prompt as a Contract (ai-3/5). Goal: Write a prompt specification with observable response criteria. Short practice: Why should the tool not fill melting from memory even if it knows a common definition?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

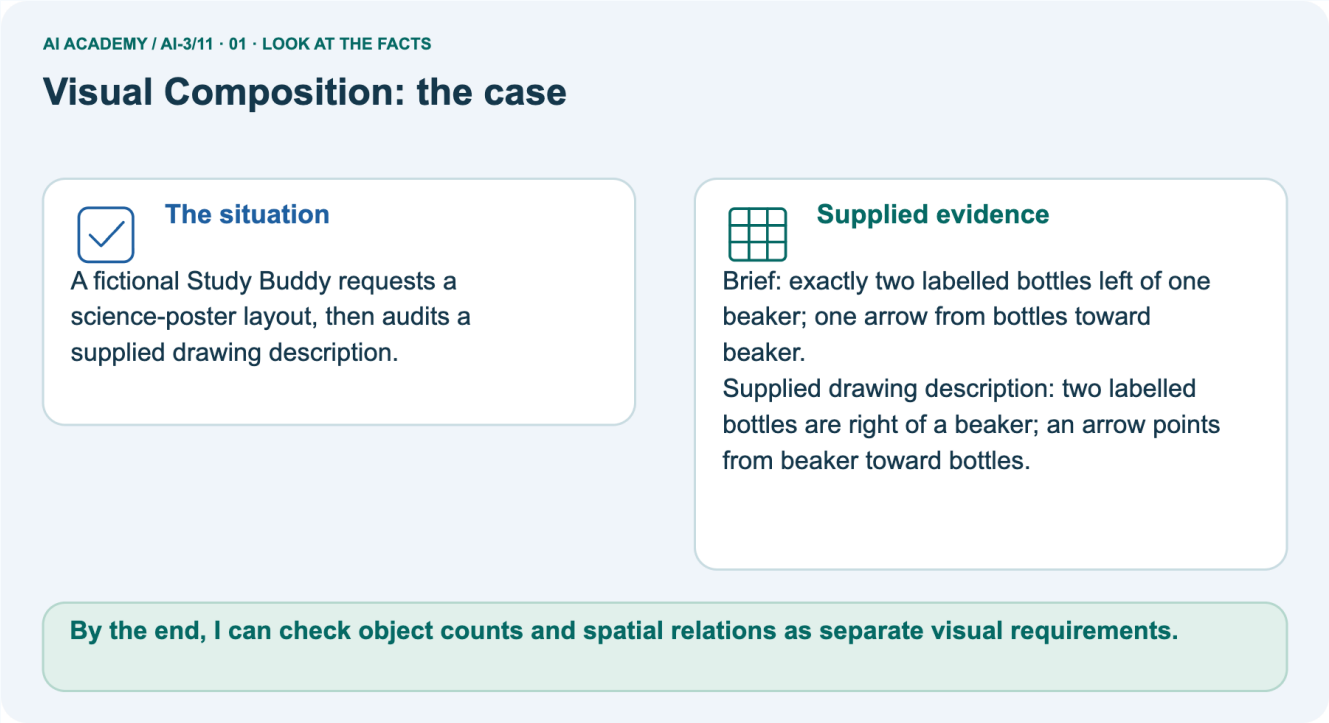
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Visual Composition picture from its labels, then record a relevant limit.

01. Visual Composition: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

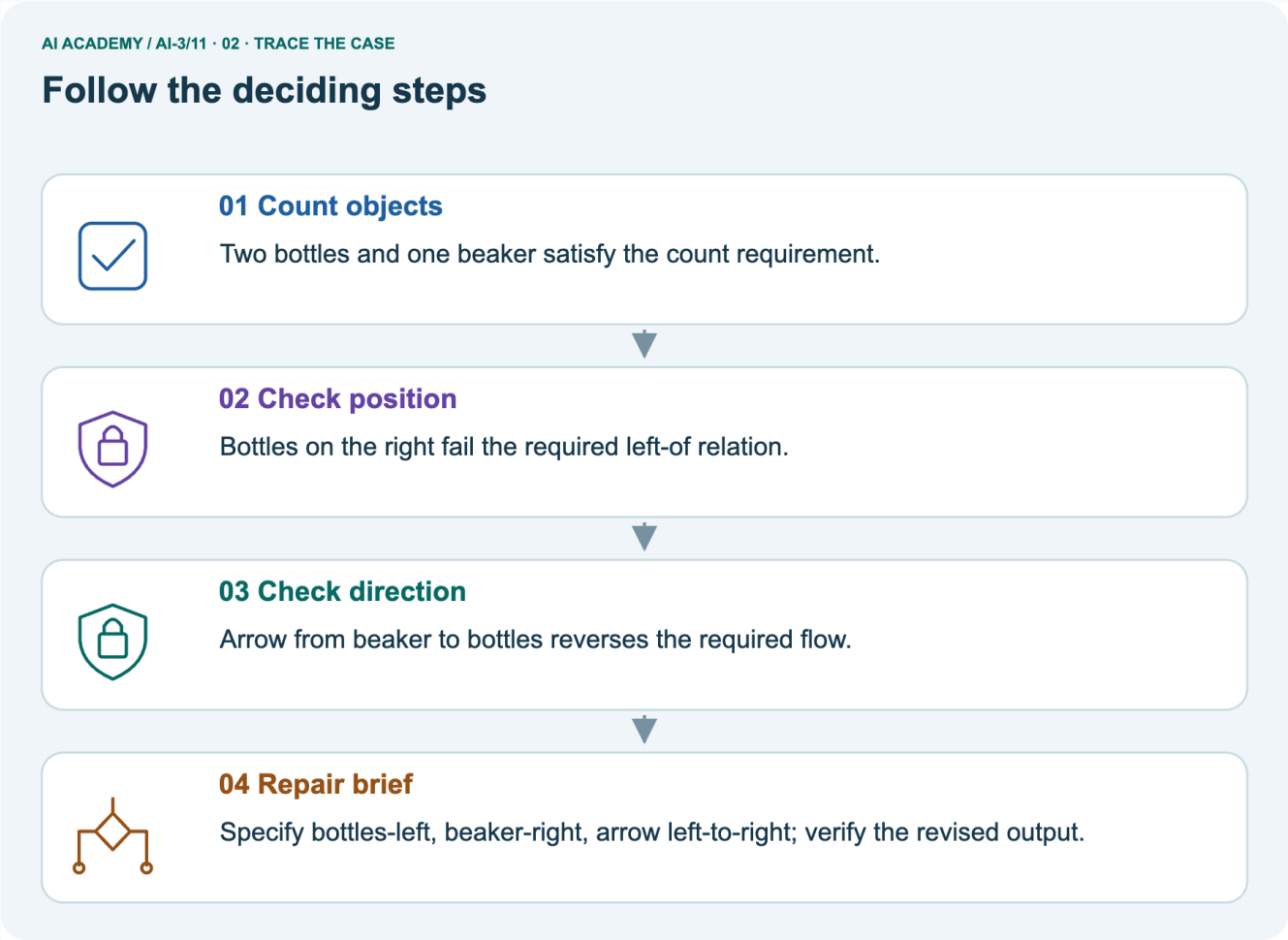
A fictional Study Buddy requests a science-poster layout, then audits a supplied drawing description. Supplied facts: Brief: exactly two labelled bottles left of one beaker; one arrow from bottles toward beaker.

Supplied drawing description: two labelled bottles are right of a beaker; an arrow points from beaker toward bottles.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



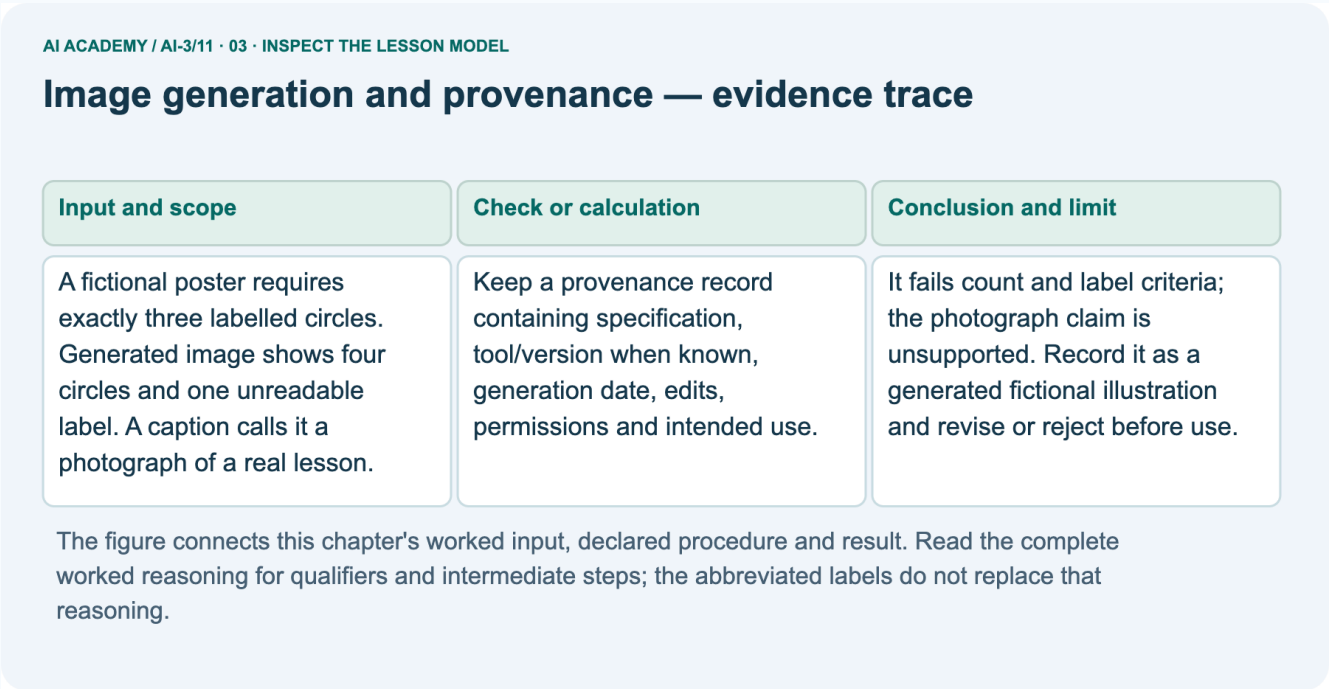
Follow the labelled picture with the matching explanation. Open the vector diagram

Count objects: Two bottles and one beaker satisfy the count requirement. → Check position: Bottles on the right fail the required left-of relation. → Check direction: Arrow from beaker to bottles reverses the required flow. → Repair brief: Specify bottles-left, beaker-right, arrow left-to-right; verify the revised **output**.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Image generation and provenance — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Image generation and **provenance** — **evidence** trace. A fictional poster requires exactly three labelled circles. Generated image shows four circles and one unreadable label.

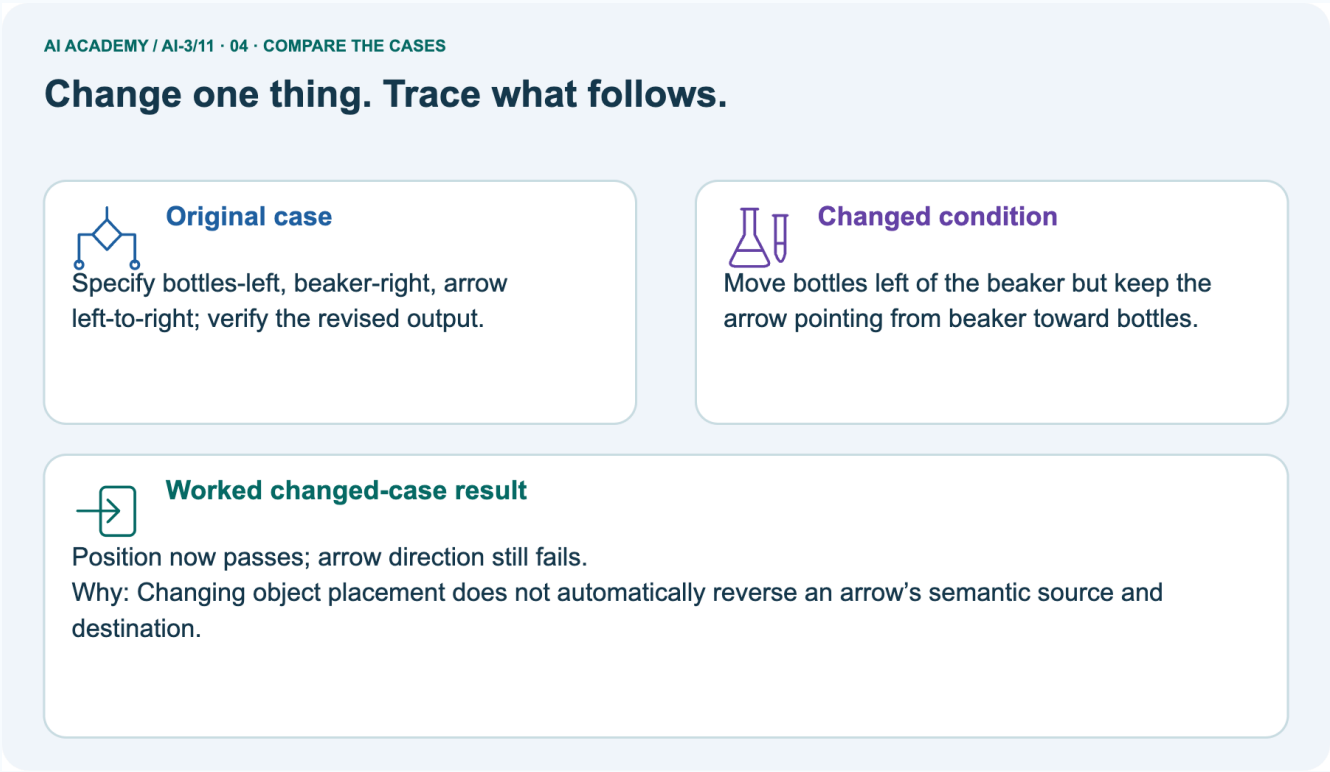
A caption calls it a photograph of a real lesson. | Keep a **provenance** record containing specification, tool/version when known, generation date, edits, permissions and intended use. | It fails count and label criteria; the photograph claim is unsupported.

Record it as a generated fictional illustration and revise or reject before use.

Read the labels and values. Explain the **relationship** the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

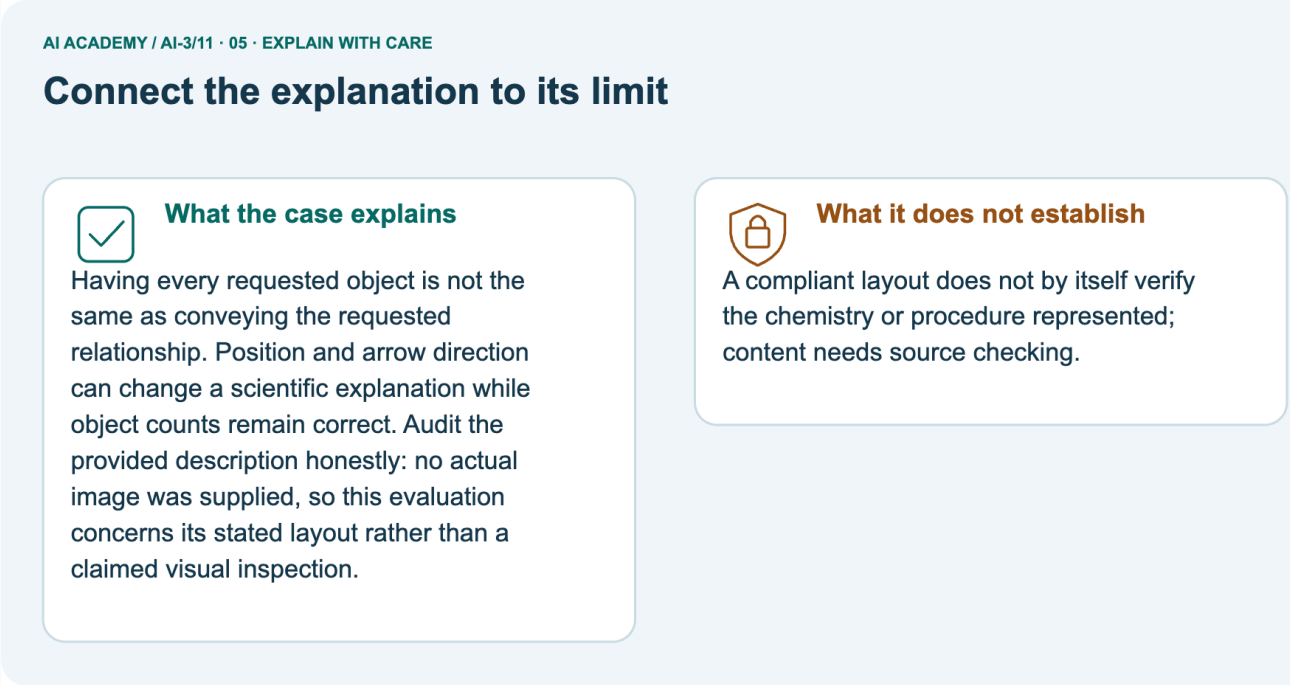
Original case: Specify bottles-left, beaker-right, arrow left-to-right; verify the revised **output**. Changed condition: Move bottles left of the beaker but keep the arrow pointing from beaker toward bottles.

Expected result: Position now passes; arrow direction still fails. Reason: Changing object placement does not automatically reverse an arrow's semantic source and destination.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Having every requested object is not the same as conveying the requested **relationship**. Position and arrow direction can change a scientific explanation while object counts remain correct.

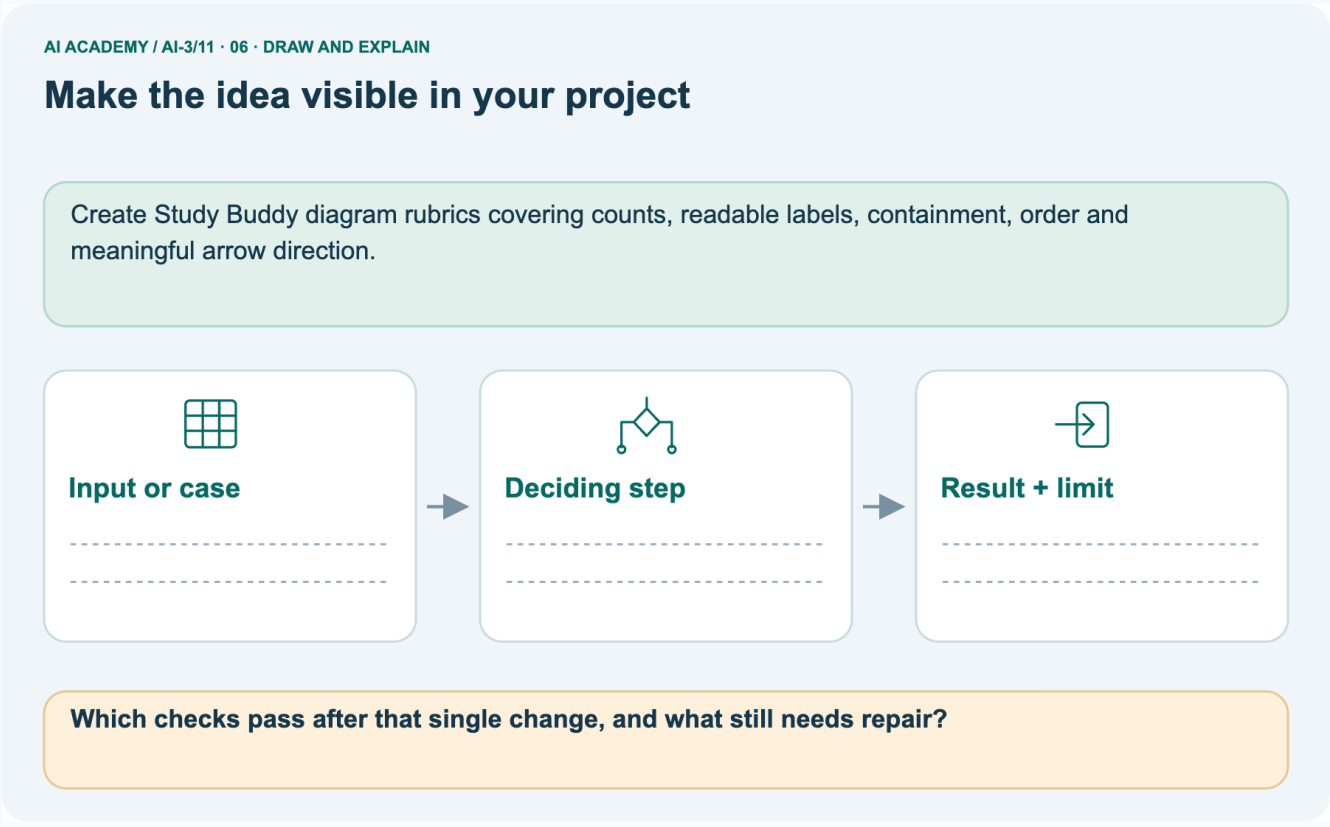
Audit the provided description honestly: no actual image was supplied, so this evaluation concerns its stated layout rather than a claimed visual inspection.

Limit: A compliant layout does not by itself verify the chemistry or procedure represented; content needs source checking.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Create Study Buddy diagram rubrics covering counts, readable labels, containment, order and meaningful arrow direction. Draw your **input**, deciding step, result and limit.

Supported question: Which checks pass after that single change, and what still needs repair?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.